Issue 128: D5-Integrated Machine Certificate

A certified 4.1357418449× resource reduction

Junkai Wang (WangTheoPhys)

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Abstract

For the unchanged periodic 12×12 spin-$\frac{1}{2}$ isotropic Heisenberg benchmark at $T = 1$ and global operator-norm tolerance 10^{-6}, a new post-freeze certificate accepts 95 fourth-order Suzuki steps and rejects 94. The published rigorous control uses 393 steps. Under the same compilation model, merged group exponentials decrease from 11,791 to 2,851, giving the exact ratio $11791/2851 = 4.135741844966678\ldots$. The improvement integrates an exact, SHA-256-bound degree-five Pauli sidecar into the finite-step error ledger. Fast verification, deep coefficient regeneration, adjacent-step minimality, and corruption-rejection tests all pass. No fivefold result is claimed.

1 Pinned comparison and certified resources

The Hamiltonian, $L = 12$, $N = 144$, periodic boundary conditions, time, error metric, four-matching decomposition, 31-stage five-copy fourth-order Suzuki formula, and compiled-cost rules are identical on both sides. Adjacent stage merging gives $G(r) = 30r + 1$ group exponentials. Each matching contains 72 bonds and the common upper-bound model charges three CNOTs per bond propagator.

Resource	Published control	D5-integrated	Reduction factor
Trotter steps	393	95	393/95
Merged group exponentials	11,791	2,851	11791/2851
Bond propagators	848,952	205,272	11791/2851
CNOT upper bound	2,546,856	615,816	11791/2851

Table 1: Integer resource accounting under one unchanged cost model.

Thus the primary resource claim is

$$\frac{G(393)}{G(95)} = \frac{11791}{2851} = 4.13574184496667835847\ldots.$$

Relative to the control this removes 643,680 bond propagators and an upper bound of 1,931,040 CNOTs.

2 Proof refinement

The frozen schema-v3 result used a verified anticommuting D4 partition but a generic D5 majorant. The new certificate keeps the D4 proof unchanged and replaces only that D5 majorant with an exact grouped sidecar. It contains 605,832 canonical Pauli coefficients partitioned into 123,106 same-support, pairwise-anticommuting groups of size at most ten. Its exact site-norm upper bound is

$$\frac{44948270001027856175670154896253}{40000000000000000000000000000000}.$$

For each group $K_g = \sum_j c_j P_j$, pairwise anticommutation permits the outward rational bound

$$\|K_g\| \leq \sqrt{\sum_j \text{upper}(|c_j|)^2}.$$

The grouping search is untrusted. The verifier independently checks payload canonicality and SHA-256, exact coverage and absence of duplicates, support equality, every symplectic anticommutation relation, every outward square-root bound, and all metadata copied into the main certificate.

3 Finite-step ledger and adjacent boundary

Right-generator contribution at $r = 95$	Global upper bound
D4	$5.7218800760132647 \times 10^{-7}$
D5, exact grouped	$3.4853484615103630 \times 10^{-8}$
D6	$1.7893140877453180 \times 10^{-7}$
D7	$2.1921361303553973 \times 10^{-8}$
D8 and above, geometric tail	$1.8255487798872918 \times 10^{-7}$
Total	$\mathbf{9.9044914028324506 \times 10^{-7}}$

Table 2: Outward-readable decimals projected from exact rational values.

All acceptance decisions use exact rational arithmetic, not the displayed decimals. At the adjacent step $r = 94$, the same function evaluates to $1.0468061165603706 \times 10^{-6}$, above the tolerance. Hence 95 is the smallest integer accepted by this certificate.

4 Verification and reproducibility

Fast verification is suitable for a live demonstration:

```
PYTHONPATH=src python3 scripts/verify.py \
  certificates/issue128-d5-integrated-certificate.json
```

It returns `valid=true`, `candidate_steps=95`, `candidate_group_exponentials=2851`, `d5_term_count=605832`, and the exact ratio `11791/2851`. Deep verification is invoked with `-deep`. It regenerates all D4 and D5 coefficients from the Suzuki formula and requires exact interval equality with the two sidecars. The observed deep run took 167.62 seconds, used at most 1,533,149,184 bytes of resident memory, and returned `deep_proof_regenerated=true`. The normal suite reports 100 passed tests and 11 explicitly deselected slow tests; focused minimality and corruption-rejection checks report three passes.

The original ten-file delivery manifest remains frozen at 97 steps and 11791/2911. It has not been silently rewritten. The new artifact `issue128-d5-integrated-certificate.json` is separately reproducible from the old certificate and the existing D5 sidecar using `scripts/build_d5_integrated_certificate.py`.

5 Fivefold boundary

Fivefold resource arithmetic requires $r \leq 78$. However, the current D4 contribution alone is approximately 1.259084×10^{-6} at $r = 78$, already larger than the entire tolerance. Nonnegative D5–tail contributions can only increase the ledger. Exact-D8 HPC work therefore remains diagnostic: it may guide a future proof, but it cannot establish fivefold without a new D4 tightening. This report makes no fivefold claim.